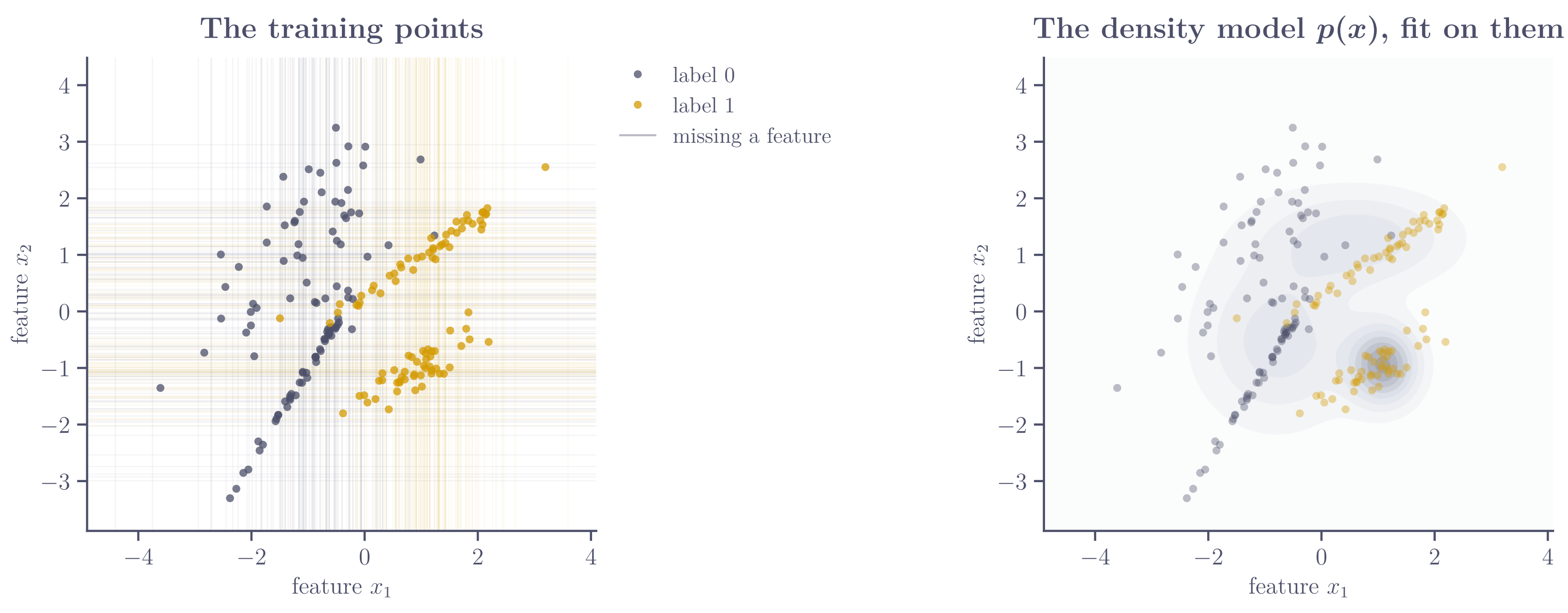
SETUP

Training points with a missing feature

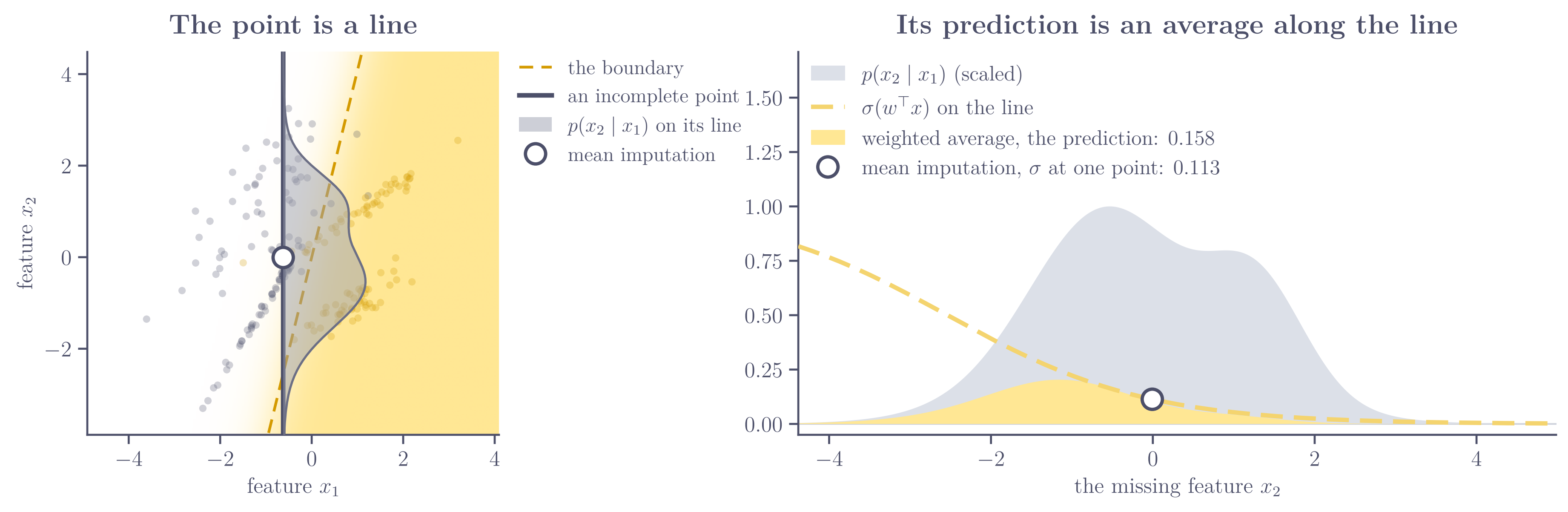

The training points

The density model $p(x)$, fit on them

600 training points with two features, yellow label 1, slate label 0. 402 of them miss one feature: they are lines, because the missing coordinate could be anywhere on them. First, a density model $p(x)$ (four Gaussians) is fit on these points, its likelihood leaves the missing features out. Then the logistic regression $\sigma(w^\top x)$ is trained: a boundary through the origin, its direction and steepness set by w.

STEP 1

An incomplete point is a line of possible points

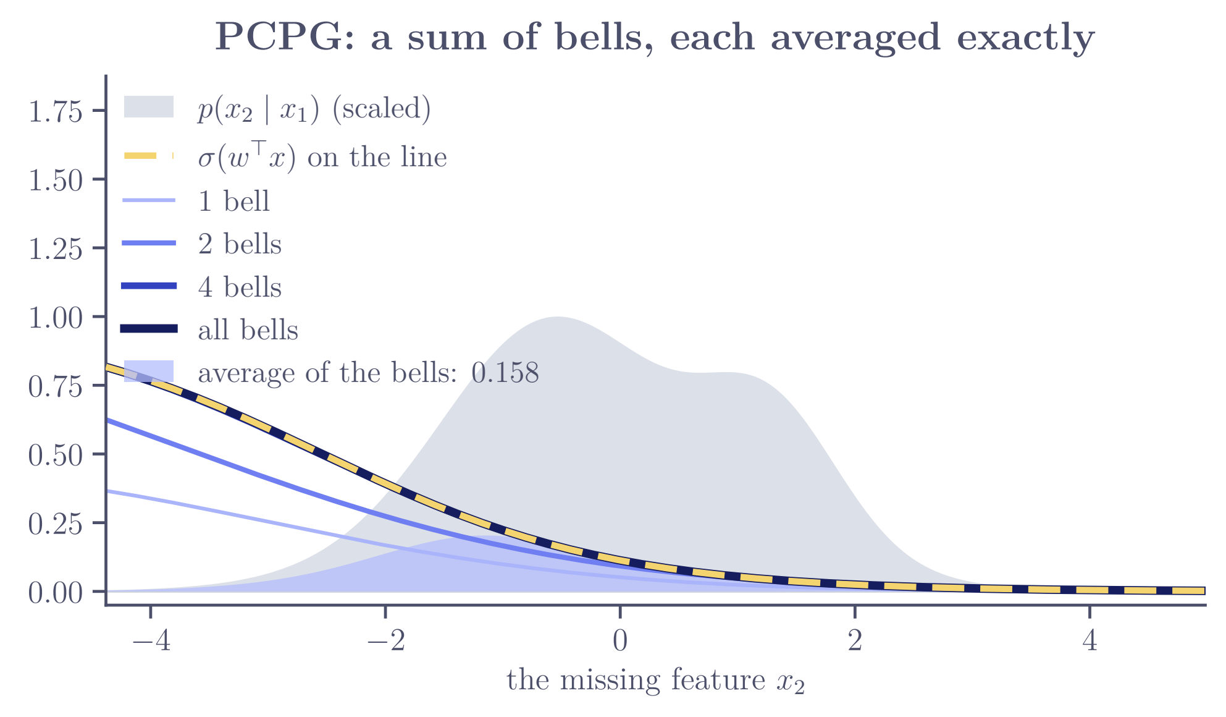

The point is a line

Its prediction is an average along the line

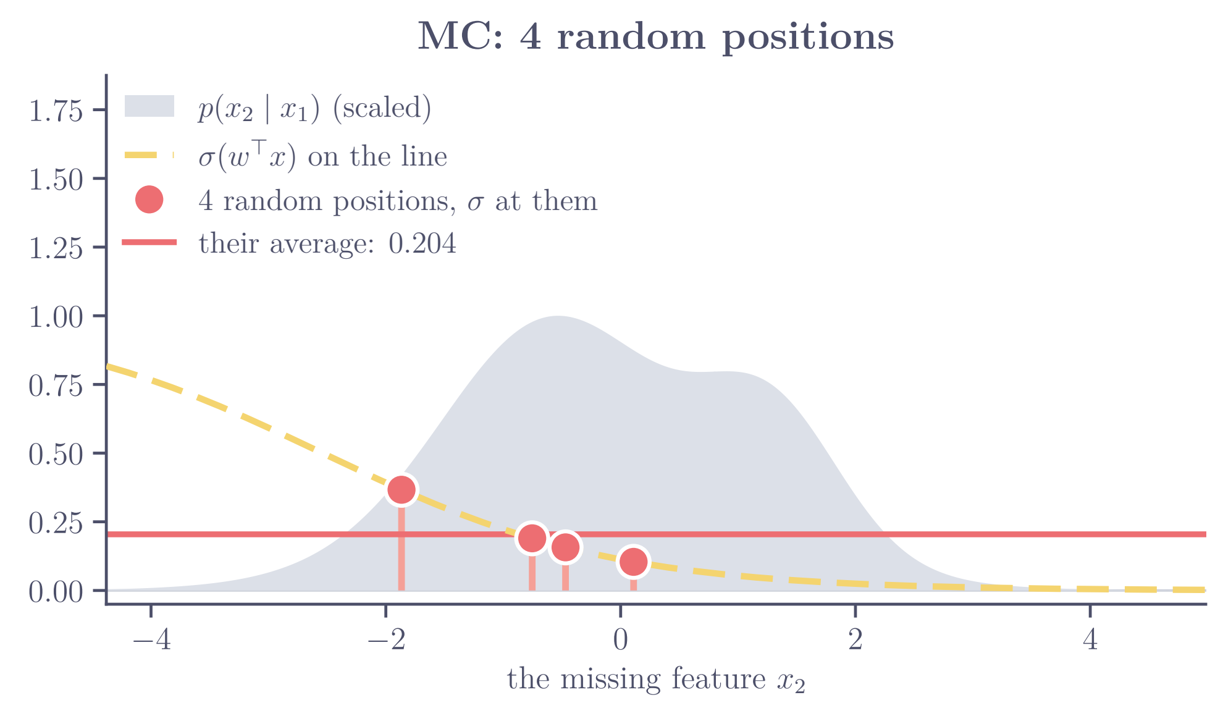
The density model says where on its line an incomplete point probably is (grey, here two bumps). Its prediction is the average of $\sigma(w^\top x)$ along the line, weighted by $p(x_2 \mid x_1)$: the yellow area, 0.158. Mean imputation takes one point instead, the mean of x_2 (white dot), and gets 0.113.

STEP 2

Two ways to compute the average


PCPG: a sum of bells, each averaged exactly

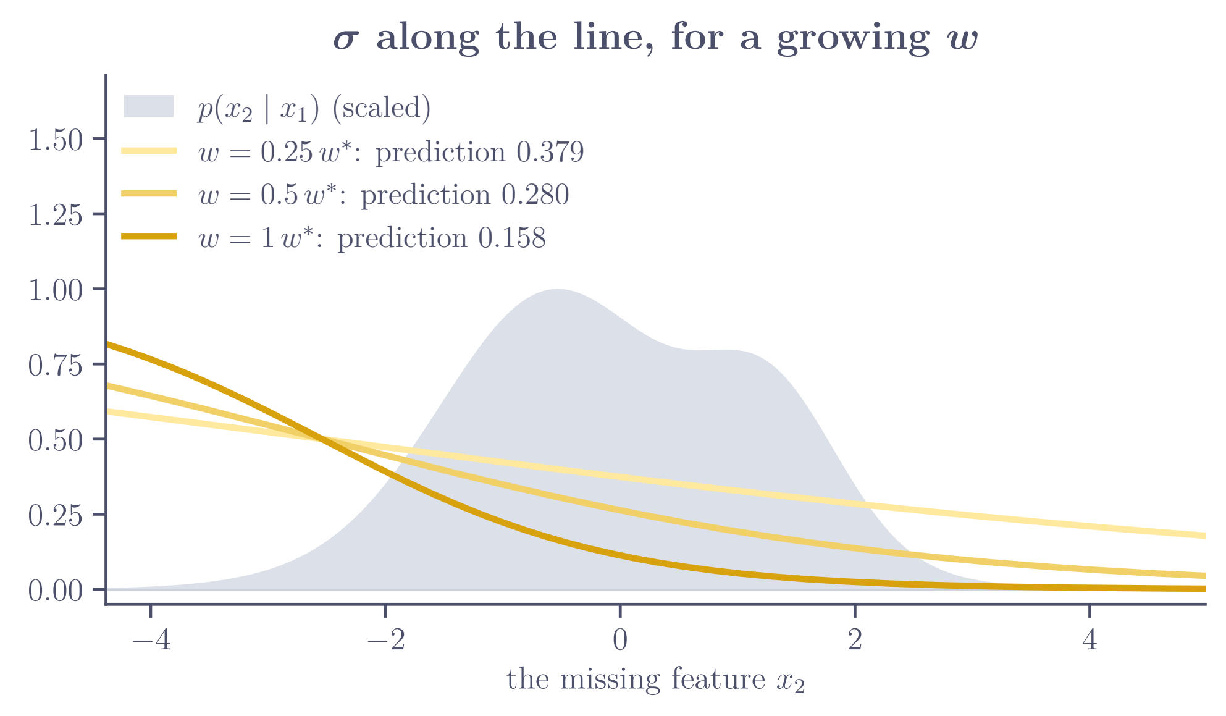
PCPG replaces σ by a sum of bells (indigo, with more bells closer to σ). The average of every bell along the line is known exactly from the characteristic function of the density model (see the PCPG step by step), so the area is added up from the bells, without trying any point on the line: 0.158.


MC: 4 random positions

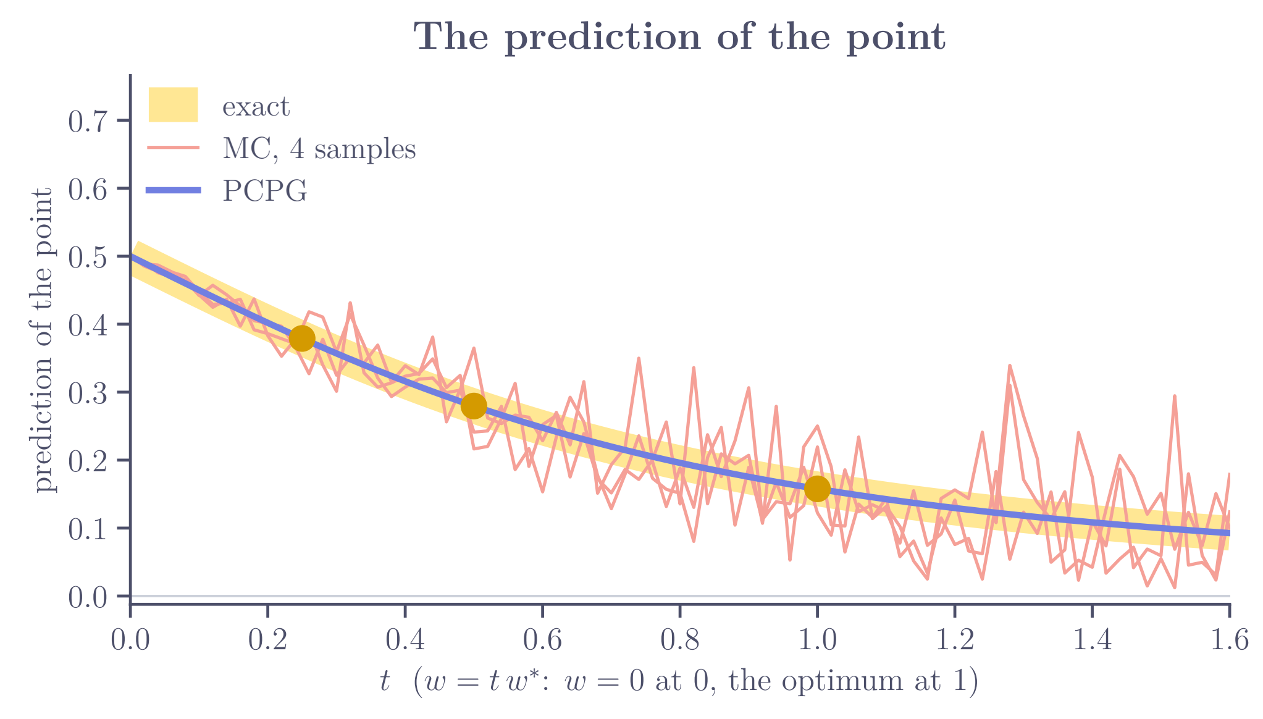
MC draws 4 random positions on the line from the density model (coral) and averages σ at them: 0.204. Other positions give another average, the result is random.

STEP 3

Training changes w, and with it the prediction


σ along the line, for a growing w

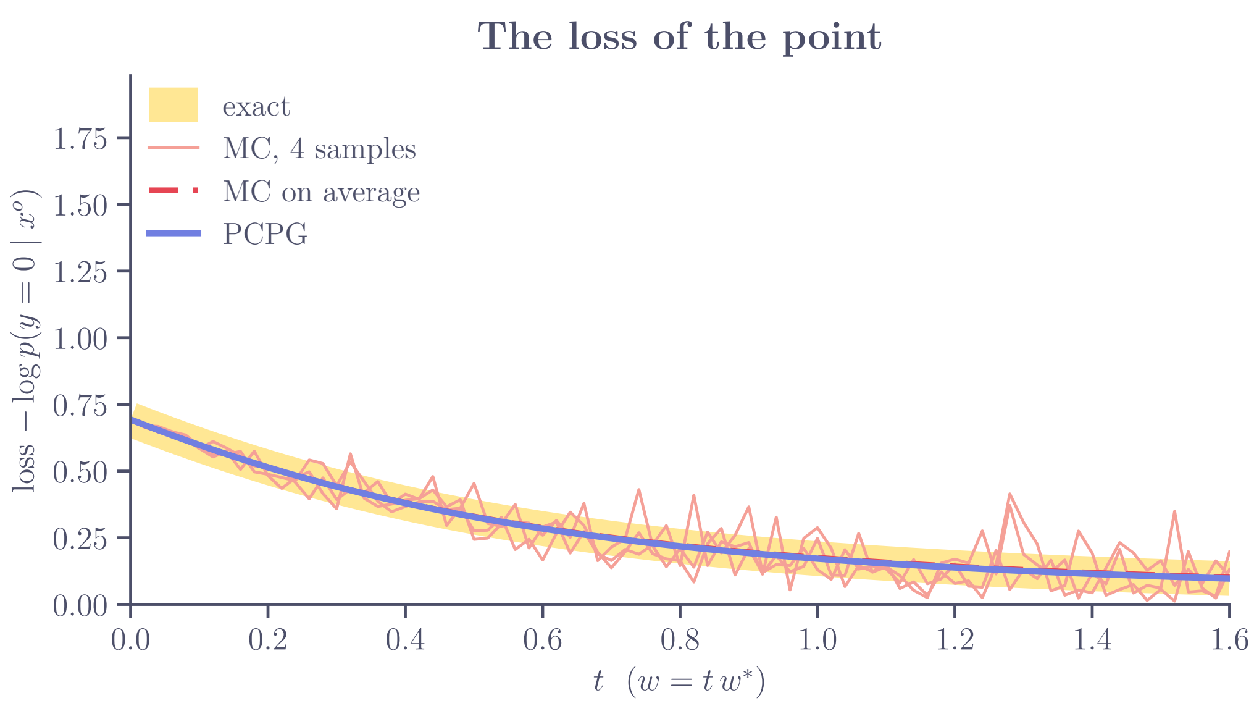
Training changes w step by step. Here w grows along a straight line from $w = 0$ to the trained w^*, $w = t\,w^*$: the longer w, the steeper σ along the line, and the area, the prediction, changes.


The prediction of the point

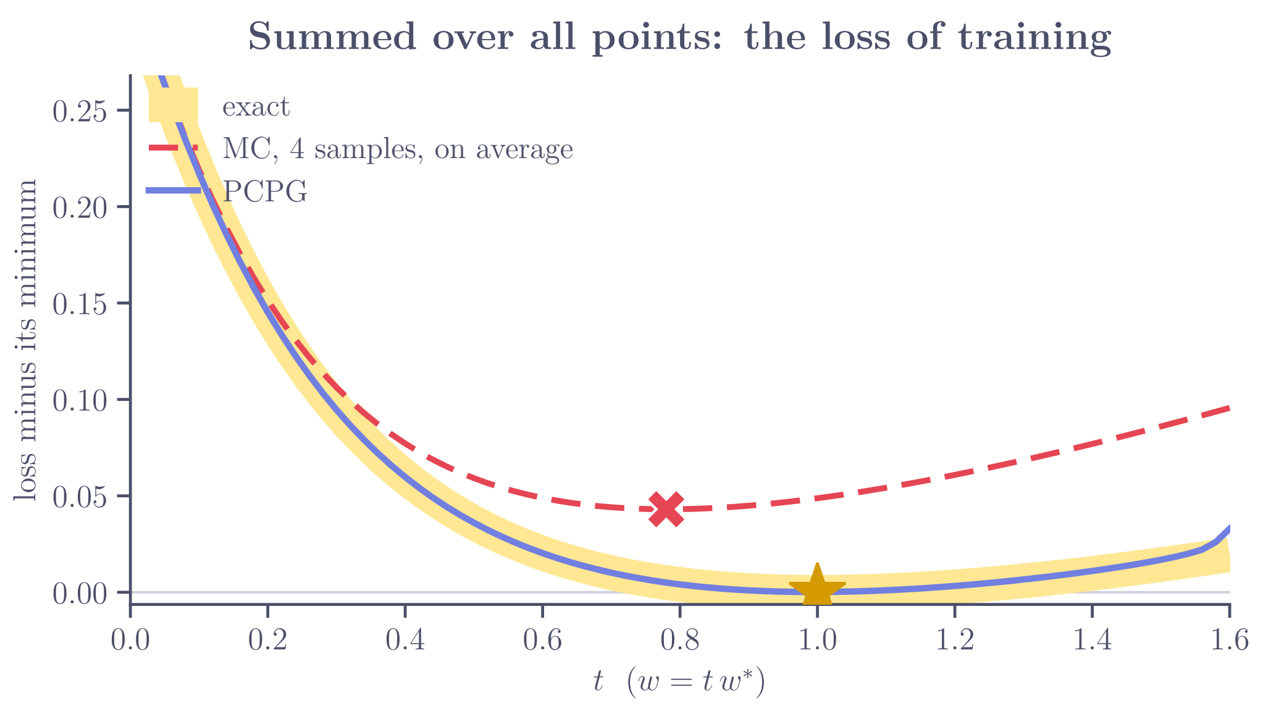
The prediction of the point as a function of t. Training needs it, and its derivative, for every w it visits. PCPG's prediction (indigo) is the exact one (yellow), a smooth function of w. MC's (4 samples, new ones at every t, three runs) jumps around it.

STEP 4

From the prediction to the loss


The loss of the point

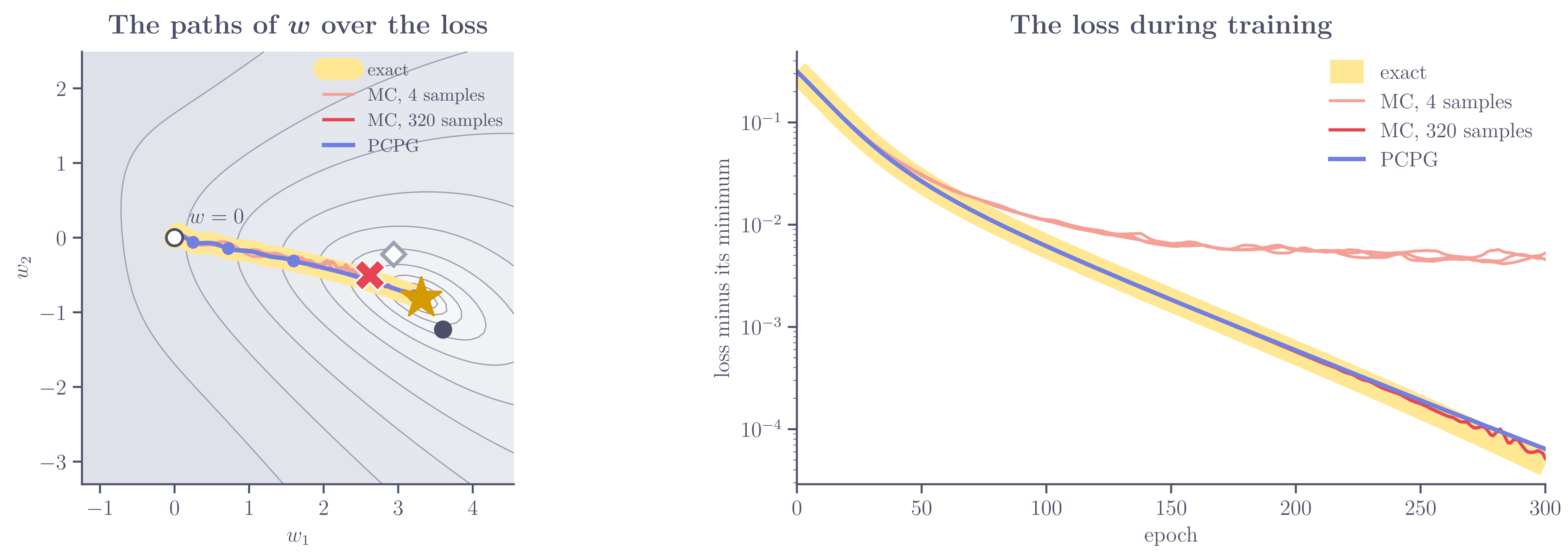
The loss of the point is $-\log$ of its prediction of its label, here $y = 0$. MC's loss is just as noisy (thin), and on average (dashed) too high: $-\log$ of a noisy prediction is too large on average (Jensen), the more the noisier the prediction.


Summed over all points: the loss of training

Summed over all points, the loss of training. PCPG's is the exact one, with its minimum at the optimum w^* ($t = 1$). MC's is too high on average, more so for a longer w, so its minimum lies at $t = 0.78$.

STEP 5

Training walks downhill

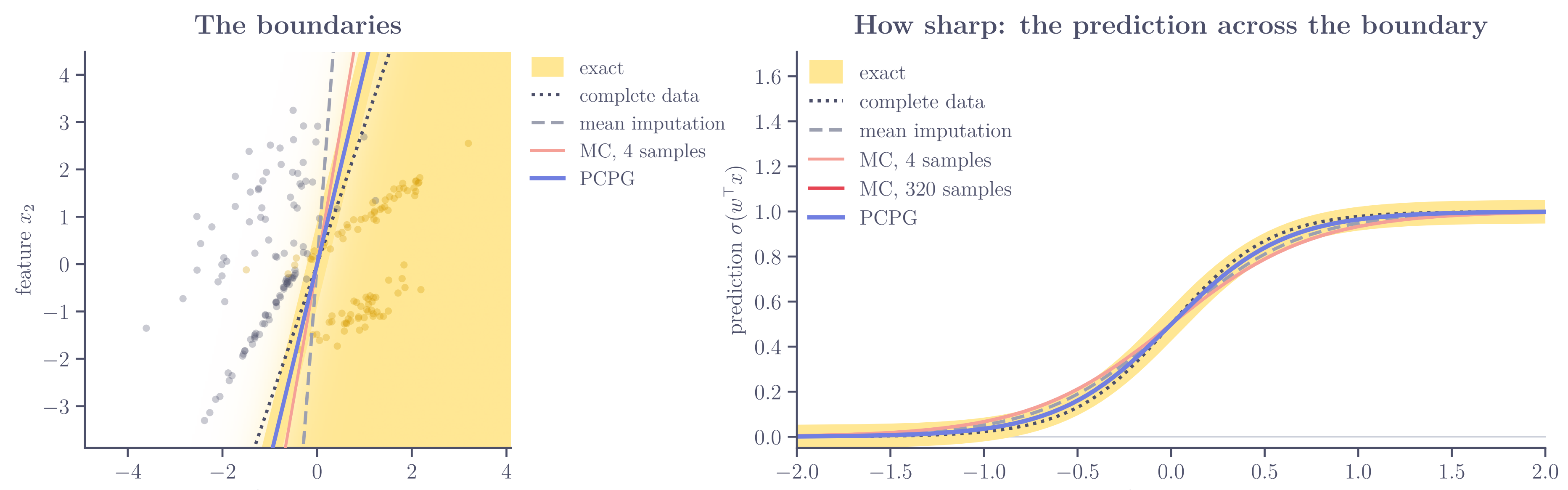

The paths of w over the loss

The loss during training

With two features, w is a point in a plane: grey the loss, lowest at w^* (star). The dot and the diamond are the classifiers on complete data and with mean imputation. Training starts at $w = 0$ and walks downhill: PCPG the same path as exact training, to w^*; MC with 4 samples to the minimum of its own loss (cross), with 320 samples near w^*. The loss during training, minus its minimum: PCPG goes down like exact training, MC with 4 samples stops far above, with 320 samples it gets close.

STEP 6

Where training ends


The boundaries

How sharp: the prediction across the boundary

The boundaries in the data: PCPG ends on the exact boundary, close to complete data, which saw all features. Mean imputation is turned by 9° towards the axis: the filled-in means lose how the two features go together. Across the boundary the prediction rises the faster, the longer w is: $\|w\| = 3.31$ for PCPG, 3.81 for complete data, 2.62 for MC with 4 samples (too unsure) and 2.95 for mean imputation.